Automated Trust Management under LLM Monitoring Uncertainty: Adversarial Risk-Aware Deployment in a Decentralized Multi-Agent Network

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ABSTRACT

Decentralized multi-agent systems increasingly rely on shared information, yet repeated interactions allow locally plausible actions to accumulate into distributed harm. We propose an automated trust-management framework that converts noisy monitor scores into directed local trust, aggregates them into a global reputation via an EigenTrust-style process, and removes persistently low-reputation agents. Building on this framework, we study an adaptive Red syndicate using shared Q-learning to farm trust, attack benign agents, smear reputations, or boost malicious teammates. We evaluate pretrust, removal threshold population, defense calibration, and Red observation regimes on attacks, Red survival and eradication, and Blue false isolation. In evaluated regimes, including pretrust anchors in threshold computation can prevent removal; excluding them restores operation but reveals an attack-false isolation trade-off. Full manipulation generally increases attacks and Red survival, while post-attack credence has limited aggregate impact despite affecting greedy action preferences. Results characterize adversarial trust dynamics under uncertain monitoring.

Code and configurations for reproducing our experiments are available at https://github.com/zong-ye-li/AutoTrust.

Keywords: *Decentralized Multi-Agent Systems, Trust Management, LLM Monitoring, Adversarial Robustness.*

1. INTRODUCTION

Networked large language model (LLM) agents are moving from isolated task execution toward persistent collaboration in which agents exchange intermediate findings, semantic representations, and memories. Such information sharing can reduce duplicated work and preserve reasoning context across agents. Recent semantic infrastructure proposals accordingly support the storage, evaluation, and transfer of cognitive state among collaborating agents [1]. However, persistent communication channels also create trust dependencies. That is, information accepted by an agent may affect later decisions by others, even if no single message appears immediately harmful. This risk is difficult to address through isolated action monitoring. Chen et al. [2] demonstrate that individually plausible subtasks can jointly produce sequentially contextual harm, while the insertion of unrelated benign subtasks can further reduce the accuracy of LLM-based monitors. External source content also blurs the boundary between data and instructions, allowing indirect prompt injection to alter an agent's behavior without direct adversarial access to its prompt [3]. When agents retain prior experiences, malicious content may persist beyond the original interaction. MemoryGraft, for example, shows that poisoned experiences can later be retrieved for semantically similar tasks and induce persistent behavioral drift [4]. These findings motivate defenses that evaluate not only individual outputs but also the trustworthiness of agents over repeated interactions.

The problem becomes more consequential when agents operate across decentralized infrastructures. Such deployment can distribute computation and reduce dependence on a single controlling service, but it also diffuses accountability and makes platform-level intervention less direct [5]. Trust management offers one possible response by aggregating local observations into network-level reputations that can guide routing or removal decisions openly. Nevertheless, an adaptive malicious group may exploit the same mechanism by behaving benignly to accumulate trust, attacking selected peers, or manipulating the reputations of other nodes. A defense must therefore limit sustained malicious activity without unnecessarily isolating benign participants. Such trust management mechanisms also benefit Internet of Medical Things (IoMT) environments, where interconnected medical devices exchange physiological data for coordinated care [6]. Sensor faults, noisy measurements, or cyber compromise may introduce unreliable information, which reputation-based trust management could help contain before it propagates through the network [7].

This study draws methodological inspiration from Wen et al. [8], particularly their use of utility-based evaluation and Bayesian credence to quantify deployment risk under model uncertainty. Their framework uses Bayesian inference at the macro-protocol level to select

among deployment protocols with different safety-usefulness profiles. We adapt these economic and probabilistic perspectives to repeated interactions among persistent nodes. Specifically, locally observed outcomes are aggregated through an EigenTrust-style reputation process [9], which subsequently informs network-level reputation and removal decisions. This setting introduces strategic interactions that are not captured by a single model-level credence, enabling malicious nodes to farm trust, manipulate peer reputations, and adapt their behavior to public reputation information and to their estimates of how targets perceive them.

We develop a simulation framework that combines uncertain LLM-monitor score distributions as denoised priors, directed local trust updates, global reputation aggregation, deviation-based node removal, and a shared Red-team Q-learning policy. The experiments vary pretrust, the population used to calibrate the removal threshold, defense strictness, and the information available to the Red policy. We evaluate the resulting interaction using cumulative delivered attacks, Red mean lifetime and complete eradication, and Blue false isolation. Fixed attack policies provide descriptive references for interpreting learned behavior. In evaluated regimes, simulations show that including pretrust anchors in threshold calibration can prevent removal, whereas excluding them restores removal but exposes an attack-false isolation trade-off; estimated post-attack credence alters greedy-action preferences without consistent aggregate gains.

2. TRUST MANAGEMENT FRAMEWORK

2.1. Network and Monitoring Setting

We consider a fixed complete network of 50 agents, the size denoted as n, five (i.e., 10%) of which belong to one malicious Red syndicate. In each epoch, every surviving agent initiates one interaction with a surviving peer selected through reputation-weighted routing. Each interaction has two evaluation directions: the recipient evaluates the initiator's request and vice versa; the resulting assessments update directed local trust. Interactions are decentralized, but a trusted Global Trust Registry (GTR) aggregates submitted trust values, publishes global reputations, and applies a fixed node removal rule. It does not inspect interaction content or run an LLM monitor.

Monitoring uncertainty is represented by offline, class conditional distributions of ordinal scores from 1 to 10 to reflect imperfect, class-dependent monitor assessments. Qwen3.7-Plus [10] was used to score labeled examples from WildJailbreak [11] for requests and FormAI-v2 [12] for generations. The simulation samples from the resulting benign- and malicious-mode score distributions; it does not call the LLM during network interactions. These fixed reference distributions represent the evaluated monitoring environment, not validated score distributions for deployment. Crucially, the Bayesian formulation is structurally agnostic to the shapes of the binned score distributions and can accommodate any normalized paired reference distributions.

2.2. Local Trust Mechanism

From a localized agent perspective, denoting $c_{ij}^{(e)} = \sigma(Z_{ij})$ as agent i's current trust in agent j at epoch e, where Z_{ij} is a latent accumulator, σ is the logistic function converting an accumulated state to bounded credence. For a score s observed at epoch $e+1$, evaluator's instantaneous credence W_B with the relevant benign- and malicious-mode reference distributions P_B and P_M:

$$W_B^{(e+1)}(c^{(e)}, s) = \frac{c^{(e)} P_B(s)}{c^{(e)} P_B(s) + (1 - c^{(e)}) P_M(s)} \quad (1)$$

The *posterior* expected utility, with risk penalty hyperparameter α for a malicious outcome, is

$$v(W_B, s) = W_B - \alpha(1 - W_B) = (1 + \alpha) W_B - \alpha \quad (2)$$

We initialize directed credence at $c^{(0)} = 0.9$ and set $\alpha = 9$. Thus, a posterior benign credence of 0.9 is the utility indifference point under the given benign and malicious payoffs. This calibration sets the relative cost of malicious evidence. Local trust evolves through a bounded update, where learning rate η limits each score's effect:

$$Z_{ij}^{(e+1)} = clip\ (Z_{ij}^{(e)} + \eta v, -3, 3), \qquad \eta = 0.1. \quad (3)$$

This update uses the Bayesian posterior of an individual observation as a utility signal for the bounded latent accumulator. The conditional *expected* utility increments under benign- and malicious-mode behavior are

$$\mu_K(c) = \sum_s P_K(s) v(W_B, s), \qquad \text{where } K \in \{B, M\}. \quad (4)$$

Their difference,

$$\Delta V(c) = \mu_B(c) - \mu_M(c), \qquad \mu_B(c) > 0 > \mu_M(c), \quad (5)$$

measures the expected utility difference between Farming and Attack, corresponding to an unclipped latent-trust increment difference of $\eta \Delta V(c)$ and provides a local accounting measure of the marginal trust cost of attack.

To relate these conditional increments to a fixed Bernoulli reference, suppose that an interaction is an Attack with probability r and Farming otherwise, while credence remains at c. Before accumulator clipping, the expected one-step shift is

$$\mathbb{E}[\Delta Z \mid c, \mathrm{r}] = \eta[\mu_B(c) - r \Delta V(c)]. \quad (6)$$

Setting the local expected change to zero gives

$$r_0(c) = \mu_B(c) / (\mu_B(c) - \mu_M(c)). \quad (7)$$

This rate defines the Bernoulli reference proportion that balances expected Farming and Attack increments at a

fixed interior credence. Under the initial utility calibration, $r_0(0.9) = 0.1$, which specifies the fixed Bernoulli policy used in the baseline comparison.

2.3. Routing Rule

GTR row-normalizes surviving agents' trust to obtain local trust matrix $C \in \mathbb{R}^{n\times n}$. Routing weight combines local trust, global reputation $\boldsymbol{t}$, and exploration constant u:

$$q_{ij} = \frac{0.60C_{ij} + 0.25\boldsymbol{t}_j + 0.15u_{ij}}{\sum_{k\in E}(0.60C_{ik} + 0.25\boldsymbol{t}_k + 0.15u_{ik})}, j \in E, \quad (8)$$

where set E contains all surviving agents, and u_{ij} is uniform over eligible surviving peers other than i. Hence, 85% of routing mass is guided by reputation, while 15% preserves stochastic interaction chances, following the damping-teleportation intuition of PageRank [13].

2.4. Global Reputation and Node Removal

Global reputation in each epoch is updated using a converged matrix derived from EigenTrust-style iteration [9]:

$$\boldsymbol{t}^{(k+1)} = (1-\beta)C^T\boldsymbol{t}^{(k)} + \beta\boldsymbol{p}, \quad \beta = 0.15, \boldsymbol{t} \in \mathbb{R}^n. \quad (9)$$

With pretrust anchors, $\mathbf{p}$ distributes mass uniformly over the verified benign anchors; otherwise, it is uniform over surviving agents. The routing exploration term and the pretrust term operate at different stages: the former affects peer selection, whereas the latter affects reputation propagation. After each epoch, the GTR sets a threshold

$$\tau = \mu_t - n\sigma_t \quad (10)$$

and permanently removes each surviving agent satisfying $t_i < \tau$. The mean and standard deviation are computed over either all or non-anchor surviving agents, dubbed threshold population.

3. ADVERSARIAL POLICIES

The learned Red-team policy is a shared tabular Q-learning policy: all Red nodes consult the same Q-table, but each decision is conditioned on the acting node's current aggregated state. The full state space contains discretized self, target, and syndicate reputation, target identity label, and estimated post-attack credence. The *reduced state* extension removes only the estimated post-attack credence coordinate. For a Blue target, the Full manipulation policy may select Farming, Attack, or Smearing; for a Red teammate, it may select Normal or Boosting. Among them, Smearing and Boosting directly set the directed latent-trust accumulator to the lower and upper clipping limits, respectively. The *No manipulation* policy retains Farming and Attack against Blue targets but removes Smearing; it selects Normal for Red-to-Red interactions. Both learned variants use the same reward structure, training protocol, and frozen greedy evaluation.

Three fixed policies provide descriptive reference points. Bernoulli-r_0 attacks a Blue target independently with probability $r_0(0.9) = 0.1$ and otherwise farms. Post-attack $c90$ and $c95$ attack only when the Red agent's predicted counterpart credence after one attack is at least 0.90 or 0.95, respectively; otherwise, they farm. All three fixed policies select Normal in Red-to-Red interactions. They use only Red-side information available at the decision point, and the counterpart credence is estimated by the fixed local trust updating rule. They do not access realized monitor scores, true latent trust, or pretrust anchor identities. Adaptive score-conditioned attacks and re-screening are outside the present evaluation; representing them would require appropriate conditional score distributions.

4. RESULTS AND DISCUSSION

Table 1 reports mean outcomes from ten independently trained, frozen Red policies per learned condition, each evaluated on 50 environment seeds; the policy is the unit of inference. It reports cumulative delivered attacks, Red mean lifetime (censored at 1,000 epochs), complete Red eradication frequency, and mean Blue false isolation at episode termination. Each cell contrasts Full/No manipulation; grey shading removes post-attack credence, and bold italics mark descriptively favorable Blue configurations. Underlining denotes fewer than 400 attacks, complete eradication above .900, or false isolation below .200; it is not a significance test. Table 2 reports fixed-policy baselines, which use seed repetition only and provide descriptive context.

4.1. Policy-Level Outcomes under Full Manipulation

Table 1 summarizes outcomes across the experimental regimes. Full manipulation generally produces more cumulative attacks and longer Red mean lifetimes than No manipulation across the tested pretrust settings, although exceptions occur, particularly in the comparison group experiment at $n = 4.0$. The differences do not consistently extend to complete Red eradication or Blue false isolation. Consequently, the following analysis focuses on Full manipulation, which permits the malicious syndicate to use both smearing and boosting strategies.

4.2. Anchor Inclusion Induces Removal Saturation

From the Blue team's perspective, we focus on defense outcomes when potential attackers use Full manipulation. In the evaluated all-alive threshold regimes with pretrust anchors, including those anchors in the threshold population results in increased dispersion and saturation. No nodes are eliminated during the 1,000-epochs horizon in these regimes, allowing Red nodes to persist and continue to deliver attack mode without eradication threat; the resulting outcomes therefore saturate.

Table 1. Policy-Level Outcomes across Blue Team's Defense and Red Team's Observation Regimes.

Threshold	Pretrust	n	Attacks (Fav. to R)	Red Mean LT (Fav. to R)	Eradication* (Fav. to B)	False Isolation (Fav. to R)
w/o Anchor (same as w/)	0	2.0	†708.5 / 11.8	†227.1 / 15.7	†.776 / 1.000	.566 / .464
		2.5	*349.9* / 32.0	117.6 / 31.7	*.902* / 1.000	.214 / .252
		3.0	*323.0* / 51.4	137.8 / 40.3	.888 / 1.000	*.184* / .096
w/o Anchor	1	2.0	†422.4 / 25.6	†151.4 / 23.6	†*.976* / 1.000	.244 / .226
		2.5	424.1 / 78.7	358.1 / 67.1	.724 / 1.000	*.103* / .174
		3.0	650.2 / 111.6	494.0 / 77.4	.572 / 1.000	*.118* / .085
	3	1.5	†924.0 / 10.3	†374.5 / 9.3	.422 / .986	.704 / .365
		2.0	*183.0* / 33.9	72.0 / 30.3	*.974* / .944	.312 / .238
		2.5	*369.4* / 51.9	113.2 / 55.0	*1.000* / .998	*.134* / .184
		3.0	*265.5* / 29.8	57.9 / 9.9	*.996* / 1.000	*.043* / .021
		3.5	837.7 / 120.7	138.0 / 69.0	.866 / 1.000	*.012* / .026
		4.0	1382.8 / 1304.6	243.1 / 272.4	.750 / .738	*.004* / .013
		1.5	†801.0 / 11.9	†208.4 / 14.4	.670 / .972	.637 / .455
		2.0	419.9 / 46.5	143.8 / 64.0	*.934* / .824	.420 / .330
		2.5	*230.8* / 14.1	63.2 / 98.5	*.980* / .864	*.100* / .231
		3.0	*199.0* / 21.7	177.7 / 2.6	.720 / 1.000	*.097* / .009
		3.5	1085.6 / 39.1	291.7 / 78.2	.686 / .884	*.019* / .021
		4.0	1089.5 / 1605.9	222.3 / 340.8	.780 / .648	*.006* / .017
w/ Anchor	1	2.0 2.5 3.0	6095.2 / 6100.5	1000.0	.000	.000
	3	1.5 2.0 2.5 3.0	6102.7 / 6103.9			

† Red team's strategy (i.e., full manipulation, post-attack est.) demonstrates dominant advantage in the marked section.
* **Eradication:** fraction of episodes with all Red eliminated; **False isolation:** fraction of benign agents falsely isolated.

Table 2. Fixed-Policy Baseline Outcomes in Full Manipulation, Pretrust=3, and Non-Anchor Regimes.

Baseline Strategy	n	Attacks (Fav. to R)	Red Mean LT (Fav. to R)	Eradication (Fav. to B)	False Isolation (Fav. to R)
Bernoulli ($\boldsymbol{p = r_0}$)	2.0	9.54	10.02		.316
	2.5	17.58	18.04	1.000	*.195*
	3.0	29.20	29.48		*.109*
†***Post-Attack*** $\boldsymbol{est.c^+ \geq 0.90}$	2.0	29.14	40.12		.565
	2.5	104.30	92.79	1.000	.319
	3.0	223.68	160.23		*.141*
Post-Attack $\boldsymbol{est.c^+ \geq 0.95}$	2.0		196.22	.440	.776
	2.5	.00	413.02	.400	.743
	3.0		811.46	.000	.270

4.3. Selected Defense Configuration

We observe a favorable exposure-isolation trade-off at pretrust $= 3$ and $n = 3.0$ under Full manipulation and non-anchor-alive thresholding. Across this displayed pretrust-3 sweep, Blue false isolation generally increases as n decreases and the removal threshold becomes stricter. We therefore further concentrate the following analysis on the curated calibration range $n = 2.0$-3.0, using the Full manipulation, pretrust-3, $n = 3.0$, non-anchor-alive regime as a favorable descriptive comparator with the full observation state.

4.4. Directionally Inconsistent Credence Estimation

A sustained attack may rely on the information available in the Red team's state space. We therefore conduct an additional ablation test that removes estimated post-attack credence from the Q-learning state. The grey-shaded rows in Table 1 show the resulting outcomes. Within the selected $n = 2.5$ and 3.0 regimes, retaining this estimate alters attack exposure, yet fails to yield advantages across other metrics and regimes and exhibits no consistent directionality across the remaining evaluation criteria.

4.5. Baseline Performance

Table 2 reports the outcomes of the three predefined baseline attack strategies in the selected defense regimes. Bernoulli attack with fixed probability $r_0(0.9) = 0.1$, a fixed Bernoulli reference calibrated to zero expected local drift at the initial credence, before clipping, yields fewer attacks than the learned Full manipulation policy. The post-attack $c^+ > 0.90$ rule delivers more than three times as many attacks as Bernoulli across the tested $n = 2.0$-3.0 regimes but yields fewer attacks than its comparator with learned policy in each displayed regime. Both active fixed policy baselines result in complete Red eradication in these regimes.

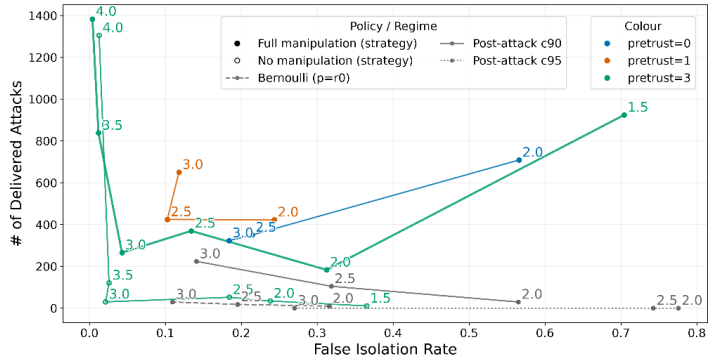


Fig. 1. Attack Exposure and Benign Isolation across Evaluated Defense and Observation Regimes.

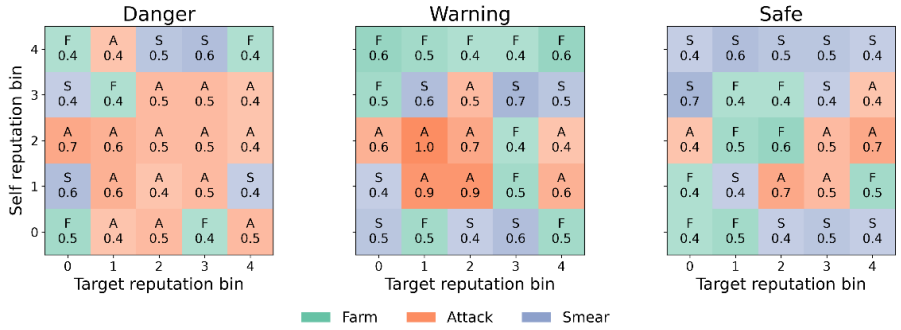


Fig. 2. Greedy Red-Team Actions by Self and Target Global Reputation and Estimated Post-Attack Credence.

In contrast, the post-attack $c^+ > 0.95$ baseline launches no attacks in the evaluated runs. This is a boundary effect of the bounded latent-trust accumulator and expected trust cost of attack specified by the hyperparameters: predicted post-attack credence does not reach the 0.95 action threshold. Zero attack exposure therefore reflects an inactive attack rule under this parameterization, rather than superior defensive performance.

4.6. Regime-Level Exposure-Isolation Trade-off

Figure 1 depicts the attack-defense outcomes using two indicators: attack exposure and false isolation. Points that share the same threshold population, pretrust setting, and policy but differ in n are connected to show within-series calibration changes; the thicker green line represents the Full manipulation, pretrust-3, non-anchor-alive series as n varies. Positions located in the upper-right or lower-left indicate worse defensive or offensive outcomes.

The all-alive regimes with pretrust anchors exhibit removal saturation, as discussed in Section 4.2, and lie far beyond the figure's displayed attack scale. They are therefore omitted from the plot. The remaining points display variation in pretrust (colours), manipulation strategy (filled or hollow circles), and fixed policy attack baseline (grey lines) relative to the selected comparator.

4.7. Role of Credence Estimation in Policy Behavior

Despite the inconsistent effect of post-attack credence estimation on metrics, its discrete bins are associated with different greedy action preferences. Figure 2 shows the modal action and the fraction of trained policies selecting it in each displayed state space cell. The slice varies self and target reputation derived from global trust, together with estimated post-attack credence: *danger*: $c^+ < 0.5$, *warning*: $0.5 \leq c^+ < 0.9$, *safe*: $c^+ \geq 0.9$. The syndicate reputation bin is held at its middle value. In this slice, safe-state cells more often favor smearing, whereas danger-state cells more often favor direct attacks.

The pattern suggests that Red policies may prefer direct attacks when they predict low post-attack credence in the target's assessment. One possible explanation is that further benign interactions would be less profitable in those states and rebuilding trust above the sustained attack threshold 0.9 is not worth farming for the Red.

Conversely, a safe post-attack credence bin means that Red predicts the target will retain relatively high trust in Red after an attack. The preference for smearing in such cells may reflect a reallocation of routing resources and interaction opportunities to other nodes, but establishing this temporal interpretation would require trajectory-level evidence.

5. LIMITATIONS AND FUTURE WORK

The reported results are specific to a complete network of 50 nodes, one malicious syndicate, and fixed monitor reference distributions. The observed exposure-isolation trade-offs should not be overinterpreted as evidence that the selected calibration is optimal under other network structures, monitor behavior, or adaptive opponents.

The Red team uses continuous expected trust updates to estimate how another node would trust it after an attack, forming a Red-side estimate of the counterpart's trust. Over repeated interactions, this estimate may become distorted, particularly when a small estimation error changes which side of an attack threshold the prediction falls on. Future work could develop methods to correct this error and provide a more reliable estimate of how the counterpart views Red, while preserving information available to Red at decision time.

The Blue team uses a static defense policy: its reputation values change, but its aggregation and deviation-driven removal rules do not adapt. One possible extension is an active defense that uses a temporal graph neural network to examine changes across epochs in the directed, normalized local trust values submitted to the GTR. Such a detector could be trained with syndicate labels and Blue false-isolation feedback, although this would require oracle supervision.

Finally, Red nodes share one Q-learning policy, while syndicate reputation compresses the group condition into a coarse state. Consequently, nodes with the same state summary choose the same greedy action, and the present policy does not represent explicit role assignments or coordinated asymmetric plans. The quantities $\mu_K(c)$ and $\Delta V(c)$ describe the expected local effect of one interaction, but future models could represent attack timing as a budget-aware discrete control problem. Such a state could combine estimated post-attack credence, an estimated trust reserve, and the anticipated value of future interaction opportunities. A controller could then select among Farming, Attack, and Smearing by comparing their immediate rewards with their continuation values. This formulation may produce alternating or burst-like attack schedules and permit coordinated group roles, rather than characterizing behavior primarily through a fixed attack proportion.

6. CONCLUSION

This study examined automated trust management under LLM-monitor uncertainty in a fixed multi-agent network with adaptive Red team behavior. The results show that threshold calibration is central to whether the removal mechanism operates. Including pretrust anchors in the calibration population prevented node removal in the evaluated regimes, whereas excluding anchors restored removal and exposed a trade-off between delivered attacks and Blue false isolation. Full manipulation generally enabled greater attack exposure and longer Red survival than the restricted policy variant. Estimated post-attack credence had limited effects on aggregate outcomes in the selected regimes, while remaining associated with distinct greedy-action preferences. These findings characterize reputation-mediated adversarial dynamics within the evaluated simulation and motivate adaptive, budget-aware extensions of trust-based defense.

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